

Data and Artificial Intelligence

Cyber Shujaa Program

Week 2 Assignment

Data Wrangling with Netflix Dataset(Kaggle)

Student Name: Hope Njoki Nyambura

Student ID: cs-eh03-25100

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1. Introduction

This assignment focused on developing hands-on experience in data wrangling using Pandas with a dataset obtained from Kaggle: [Netflix Movies and TV Shows](#).

The task provided an opportunity to practice loading, exploring, cleaning, structuring, and validating data before exporting it into a reusable format.

2. Objectives

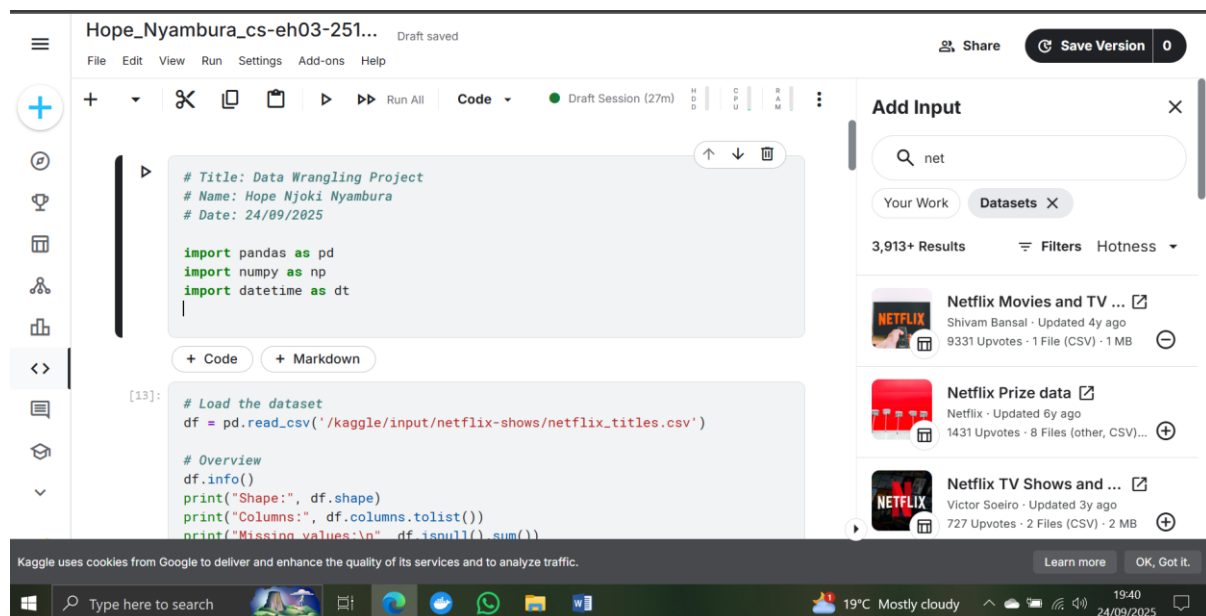
The main objectives of this assignment were:

- To load and explore the Netflix dataset using Pandas.
- To assess data types, missing values, and potential quality issues.
- To clean the dataset by handling duplicates, missing values, and formatting inconsistencies.
- To transform and enrich the dataset e.g. splitting duration into numeric and unit.
- To validate the final dataset by ensuring consistency, completeness, and logical accuracy.
- To export the cleaned dataset to CSV for future analysis or visualization.

3. Tasks Completed

Step 1: Import Libraries

Imported Pandas, Numpy, and Datetime libraries.



Step 2: Load Dataset

Loaded the Netflix dataset from Kaggle into a Pandas DataFrame.

The screenshot shows a Kaggle notebook interface. The top bar indicates the notebook is titled 'Hope_Nyambura_cs-eh03-251...' and is a 'Draft saved'. The left sidebar contains icons for file management, code execution, and other notebook features. The main area displays a code cell with the following Python code:

```
# Load the dataset
df = pd.read_csv('/kaggle/input/netflix-shows/netflix_titles.csv')

# Overview
df.info()
print("Shape:", df.shape)
print("Columns:", df.columns.tolist())
print("Missing values:\n", df.isnull().sum())
print("Duplicates:", df.duplicated().sum())
```

Below the code, the output of the `df.info()` command is shown, indicating the DataFrame has 8807 entries and 12 columns. The columns and their data types are listed:

#	Column	Non-Null Count	Dtype
0	show_id	8807 non-null	object
1	type	8807 non-null	object
2	title	8807 non-null	object
3	director	6173 non-null	object
4	cast	7982 non-null	object
5	country	7976 non-null	object

The right sidebar shows the 'Add Input' section with a search bar containing 'net'. It lists several datasets, including 'Netflix Movies and TV ...' by Shivam Bansal, 'Netflix Prize data' by Netflix, and 'Netflix TV Shows and ...' by Victor Soeiro.

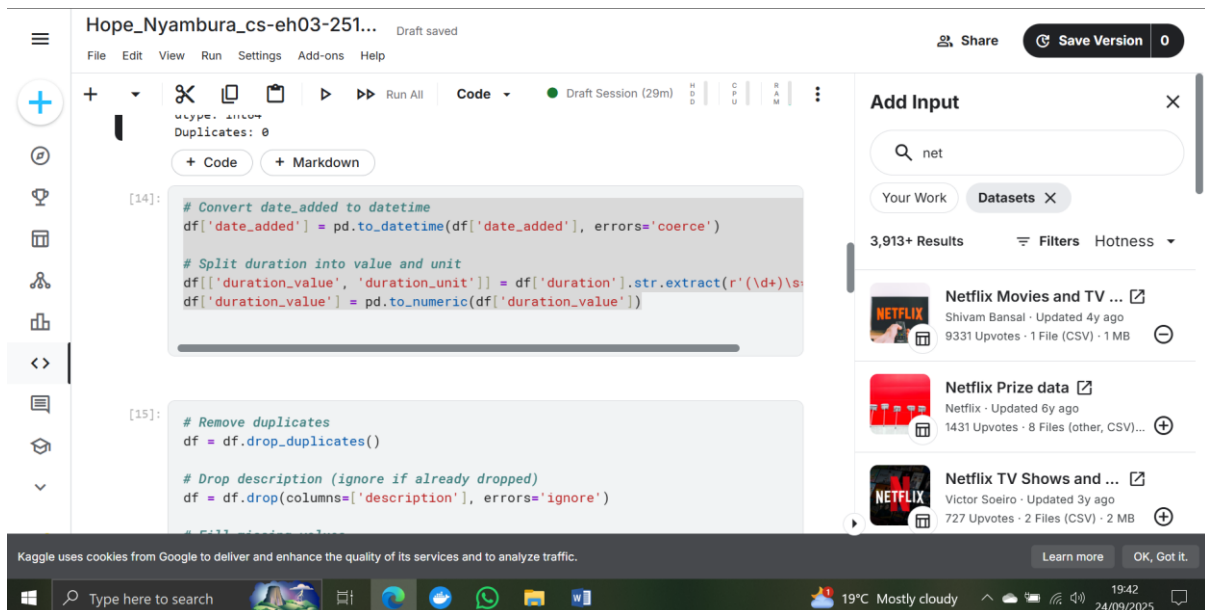
Step 3: Data Discovery

- Checked number of rows and columns.
- Inspected column names and data types.
- Assessed missing values and duplicate rows.

This screenshot is identical to the one above, showing the same Kaggle notebook interface with the code to load the Netflix dataset and the resulting DataFrame information.

Step 4: Structuring the Data

- Converted date_added column to datetime.
- Split duration into duration_value and duration_unit.



```
Hope_Nyambura_cs-eh03-251... Draft saved
File Edit View Run Settings Add-ons Help

+ Code + Markdown

[14]:
# Convert date_added to datetime
df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce')

# Split duration into value and unit
df[['duration_value', 'duration_unit']] = df['duration'].str.extract(r'(\d+)\s(\w+)')
df['duration_value'] = pd.to_numeric(df['duration_value'])

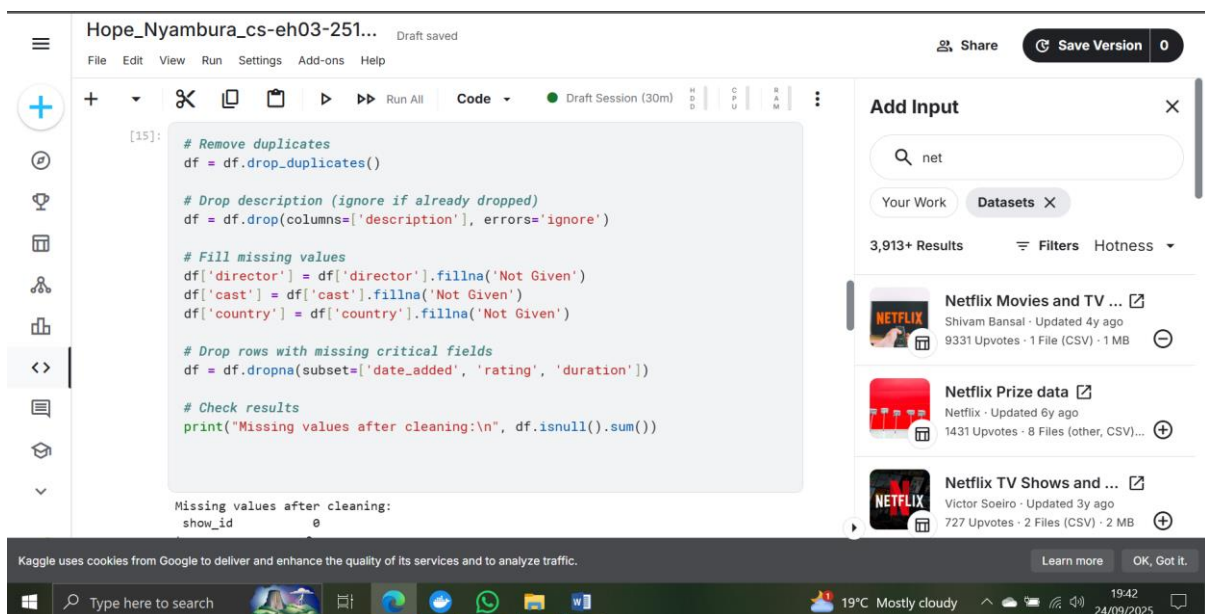
[15]:
# Remove duplicates
df = df.drop_duplicates()

# Drop description (ignore if already dropped)
df = df.drop(columns=['description'], errors='ignore')
```

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Step 5: Cleaning the Data

- Removed duplicates.
- Dropped description column.
- Filled missing values (director, cast, country) with "Not Given".
- Dropped rows missing critical values (date_added, rating, duration).



```
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[15]:
# Remove duplicates
df = df.drop_duplicates()

# Drop description (ignore if already dropped)
df = df.drop(columns=['description'], errors='ignore')

# Fill missing values
df['director'] = df['director'].fillna('Not Given')
df['cast'] = df['cast'].fillna('Not Given')
df['country'] = df['country'].fillna('Not Given')

# Drop rows with missing critical fields
df = df.dropna(subset=['date_added', 'rating', 'duration'])

# Check results
print("Missing values after cleaning:\n", df.isnull().sum())

Missing values after cleaning:
show_id      0
```

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Step 6: Validation

- Ensured date_added is datetime and duration_value is numeric.
- Checked for inconsistencies where date_added < release_year and removed 14 rows.
- Confirmed earliest release year is 1925 and latest is 2021.
- Randomly sampled rows to visually confirm cleaning accuracy.

The screenshot shows a Kaggle Notebook interface for a session titled 'Hope_Nyambura_cs-eh03-251...'. The code in the cell [16] identifies inconsistent rows where the date added is less than the release year and removes them. The output shows 14 inconsistent rows were removed. The code in cell [17] resets the index and saves the cleaned dataset to a CSV file.

```
[16]: # Find records where added date < release year
invalid = df[df['date_added'].dt.year < df['release_year']]
print("Inconsistent rows:", invalid.shape[0])

# Drop inconsistencies if necessary
df = df[~(df['date_added'].dt.year < df['release_year'])]

Inconsistent rows: 14
+ Code + Markdown

[17]: # Reset index
df = df.reset_index(drop=True)

# Save cleaned dataset
df.to_csv('/kaggle/working/cleaned_netflix.csv', index=False)
```

The right sidebar shows the 'Add Input' section with a search for 'net' and a list of datasets including 'Netflix Movies and TV ...', 'Netflix Prize data', and 'Netflix TV Shows and ...'.

Step 7: Export Final Dataset

- Reset DataFrame index.
- Exported cleaned dataset to CSV: cleaned_netflix.csv.

The screenshot shows the same Kaggle Notebook interface. The code in the cell [17] is highlighted, showing the reset index and save to CSV steps. Below it, a new cell [18] is added with code to confirm data types, print the earliest and latest release years, and perform a visual spot-check by printing a sample of 5 rows.

```
[18]: print(df.dtypes) # confirm types
print("Earliest release year:", df['release_year'].min())
print("Latest release year:", df['release_year'].max())
print(df.sample(5)) # visual spot-check
```

The output of the code shows the data types for each column: show_id (object), type (object), title (object), director (object), cast (object), country (object), and date_added (datetime64[ns]). The earliest release year is 1925 and the latest is 2021. A sample of 5 rows is printed.

show_id	type	title	director	cast	country	date_added
1	TV Show	The Big Bang Theory	Sheldon Cooper	Jim Parsons	USA	2007-09-24
2	TV Show	The Mindy Project	Mindy Kaling	Mindy Kaling	USA	2012-09-24
3	TV Show	The Normal Heart	Rob Marshall	Frank Whaley	USA	2014-09-24
4	TV Show	The Mindy Project	Mindy Kaling	Mindy Kaling	USA	2012-09-24
5	TV Show	The Mindy Project	Mindy Kaling	Mindy Kaling	USA	2012-09-24

4. Conclusion

This assignment provided valuable practice in end-to-end data wrangling using Pandas. I successfully cleaned and structured the Netflix dataset, resolved missing values, and validated the results. The cleaned dataset is now consistent, complete, and ready for further analysis or visualization.

5. Link to Kaggle Notebook

[<https://www.kaggle.com/code/defhopeeee/hope-nyambura-cs-eh03-25100>]